

EC 551 Advanced Digital Design with Verilog and FPGAs

Fall 2026, Prof. Douglas Densmore

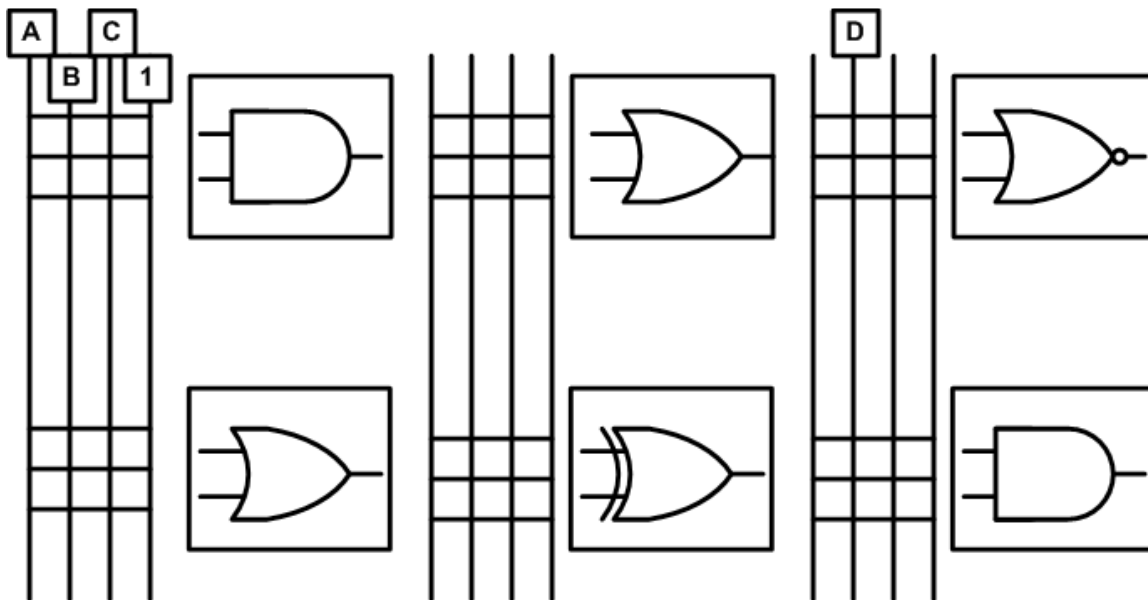
Homework #2

Due: 10/20

1. Find a comparison chart of FPGAs (e.g., within product families is fine; Altera is an older FPGA maker acquired by Intel in 2015 that you might want to use). Highlight four differences between devices and discuss how these might be either beneficial for a design or detrimental to a design. Discuss these in terms of performance, power, and cost. In addition, briefly discuss the metrics by which the FPGAs are compared (e.g. the columns in the chart perhaps).
2. Draw a schematic showing what the following Verilog code is describing.

```
reg [7:0] O, mix, Rum;  
reg [2:0] CB;  
reg A, C, D, E;  
assign O = A? {{4{C}}, mix[3:1], D^E} : {8{Rum[CB]}};
```

3. a. Connect (draw lines from gate inputs and outputs and/or add dark circles on the switch fabric) the primary inputs, gate inputs, and gate outputs to the provided FPGA switch fabric to realize the function: $(A+BC)D$



b. Assuming this design has only six 2-input LUTs and the six illustrated 4x4 crossbars, how many bits of SRAM would be needed to store connection and LUT information? (show work, state assumptions)

c. How realistic is this FPGA? What is it missing (if anything)? How would you make this better? What would be some challenges of making this FPGA?

4. Create a circuit that counts the number of 1's in a 4-bit value. The input should be 4-bits wide and the output should be a 3-bit binary number representing the value.

a. Write a structural Verilog description of the design that synthesizes.

b. Write a behavioral Verilog description of the design that will synthesize.

c. Write a behavioral Verilog description of the design that synthesizes AND the resultant circuit is different than part b.

d. Write a Verilog description that WILL NOT synthesize. Show the error messages/result that indicates that it will not synthesize. Explain why these occur.

Provide verilog code, schematics, and relevant synthesis messages for all four parts.

5. Consider a 4-bit counter. This will be a sequential logic circuit (i.e. FSM). The circuit should have a reset signal. This should set it to the initial output state of 4'b0000. The only other input should be a clock signal. Each clock tick should advance the counter. The counter should reset after 4'b1111.

a. Write a structural Verilog description of the design that synthesizes.

b. Write a behavioral Verilog description of the design that will synthesize.

c. Write a behavioral Verilog description of the design that synthesizes AND the resultant circuit is different than part b.

d. Write a Verilog description that WILL NOT synthesize. Show the error messages/result that indicates that it will not synthesize. Explain why these occur.

Provide verilog code, schematics, and relevant synthesis messages for all four parts.